



1.3 Exponents and logarithms

Number and algebra · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Logarithms are the inverse of exponentiation: $\log_a(x)=y \Leftrightarrow a^y=x$ ($a>0$, $a\neq 1$, $x>0$). Key laws: $\log(ab)=\log a+\log b$, $\log(a/b)=\log a-\log b$, $\log(a^n)=n \log a$, and the change of base formula $\log_a(b)=\ln b/\ln a$. Exponential equations of the form $a^x=k$ are solved by taking logarithms of both sides.

Worked example: follow the reasoning

Solve $3^{(2x+1)}=50$ for x , correct to 3 significant figures.

Solution: Take $\ln$ of both sides: $(2x+1)\ln 3=\ln 50 \rightarrow 2x+1 = \ln 50/\ln 3 = 3.9120/1.0986 = 3.561$. So $2x = 2.561$, giving $x = 1.28$ (3 sf).

Quick check · multiple choice

Simplify $\log_2(32)$.

- A 4
- B 5
- C 6
- D 16

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 The intensity of light passing through a filter decreases according to $I = I_0 \times 0.85^d$, where d is the thickness of the filter in mm. (a) Find the percentage of light remaining after passing through a 5 mm filter. (b) Find the thickness required to reduce the light intensity to 10% of I_0 , giving your answer to 2 decimal places. [5]

Final answer: _____



Answers and explanation

Question 1 [5]

(a) $0.85^5 = 0.4437$, so about 44.4% of light remains. (b) $0.85^d = 0.10 \rightarrow d \ln 0.85 = \ln 0.10 \rightarrow d = \ln 0.10 / \ln 0.85 = (-2.3026) / (-0.1625) \approx 14.17$ mm.

Quick check answer

B. $2^5=32$, so $\log_2(32)=5$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AI subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.